

AK-KUDUK, Kazakhstan

WMO: 382320

Lat: 42.9713N

Lon: 54.0945E

Elev: 256

StdP: 14.56

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.6	4.7	-7.5	3.7	3.1	-2.9	4.7	8.3	23.3	29.5	21.8	30.3	4.3	20	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	105.0	67.1	102.2	66.5	99.5	65.8	71.7	93.8	70.0	92.4	68.4	91.6	9.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.2	80.2	62.1	84.2	79.5	59.8	77.5	78.6	35.6	92.9	34.1	92.1	32.8	91.3	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	21.4	18.9	-5.8	109.7	9.0	3.2	-12.2	112.0	-17.5	113.8	-22.5	115.6	-29.0	117.9
			-6.0	74.9	8.7	3.0	-12.3	77.1	-17.4	78.9	-22.3	80.6	-28.6	82.8
				DB										
				WB										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	26.3	30.1	42.8	57.1	70.5	81.0	85.9	83.0	69.4	54.3	39.9	30.0
	DBStd	22.61	10.21	10.12	8.38	8.93	7.73	6.62	5.86	6.42	7.31	8.50	9.04	9.06
	HDD50	2569	735	559	250	33	0	0	0	0	0	54	318	620
	HDD65	5389	1200	978	689	266	36	1	0	0	37	343	754	1085
	CDD50	4760	0	1	27	246	637	931	1114	1022	582	186	14	0
	CDD65	2106	0	0	1	29	208	482	649	558	169	10	0	0
	CDH74	28847	0	0	11	359	2366	6590	9562	7837	1966	156	0	0
	CDH80	16445	0	0	2	102	1029	3793	5942	4719	834	24	0	0
Wind	WSAvg	7.8	6.9	7.8	8.9	9.0	8.5	9.0	8.1	7.3	7.3	7.1	6.7	7.2
Precipitation	PrecAvg	5.7	0.4	0.3	0.8	0.9	0.7	0.4	0.3	0.1	0.4	0.4	0.5	0.5
	PrecMax	12.6	0.9	1.7	2.8	5.4	2.4	2.0	2.5	0.6	2.0	1.1	1.2	1.3
	PrecMin	2.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	0.2	0.4	0.7	1.1	0.7	0.5	0.5	0.2	0.4	0.4	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	57.4	73.7	88.6	97.7	105.9	109.2	108.3	99.8	83.2	67.2	53.2
		MCWB	43.4	45.8	53.4	60.3	63.5	66.6	67.5	67.5	65.2	58.7	53.5	44.4
	2%	DB	43.8	51.0	66.6	82.7	93.4	102.2	105.6	104.4	92.4	78.4	61.9	47.3
		MCWB	40.1	43.4	50.4	58.6	62.1	65.6	67.6	66.7	62.3	57.6	51.0	42.5
	5%	DB	40.6	46.1	61.1	78.1	89.5	99.3	102.7	101.3	88.6	74.2	57.1	44.0
		MCWB	37.8	40.3	47.7	56.8	61.3	64.8	67.2	66.2	61.1	55.5	48.0	40.3
	10%	DB	37.8	42.4	56.7	73.2	86.1	96.2	99.8	98.2	84.6	69.8	52.6	41.1
		MCWB	35.5	37.8	45.6	54.4	60.8	64.3	66.6	65.4	59.7	53.6	46.0	38.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	47.2	54.7	62.9	68.2	72.4	74.6	75.1	68.9	63.7	56.3	46.2
		MCDB	48.7	55.4	69.4	80.8	86.0	92.7	95.5	92.8	88.4	74.9	63.5	50.1
	2%	WB	40.9	44.1	51.9	60.5	65.5	69.8	72.1	71.0	65.6	60.2	52.6	43.4
		MCDB	43.6	49.9	63.9	77.3	83.2	91.9	94.2	93.7	84.2	72.1	59.6	46.7
	5%	WB	38.0	41.0	49.0	58.4	63.9	68.0	70.5	68.9	63.3	57.8	49.3	40.8
		MCDB	40.1	45.4	60.0	73.8	82.2	90.7	92.9	92.5	82.9	70.3	55.1	43.5
	10%	WB	35.7	38.2	46.5	56.3	62.4	66.4	69.0	67.2	61.5	55.4	46.6	38.5
		MCDB	37.8	41.9	55.0	70.9	81.2	89.3	92.1	92.0	80.4	67.3	52.0	40.7
Mean Daily Temperature Range	5% DB	MDBR	13.3	16.0	19.6	21.8	22.3	23.5	22.8	24.6	24.9	23.1	16.7	13.3
		MCDBR	14.3	21.1	27.7	28.3	26.4	27.5	26.1	29.0	29.7	29.7	23.9	17.2
		MCWBR	11.0	13.8	14.3	12.1	9.8	9.8	8.9	11.0	13.1	15.3	14.8	13.0
	5% WB	MCDBR	13.6	19.3	25.4	25.3	23.3	23.5	22.3	24.4	25.5	24.9	20.5	15.2
		MCWBR	10.9	13.1	13.7	11.6	9.6	10.1	9.7	11.1	12.7	14.6	13.5	12.0
Clear-Sky Solar Irradiance	taub	0.388	0.413	0.465	0.536	0.523	0.505	0.507	0.484	0.455	0.439	0.401	0.391	
	taud	2.041	1.995	1.908	1.773	1.851	1.919	1.926	1.968	1.995	2.006	2.067	2.044	
	Ebn at Noon	220	238	241	234	243	247	245	246	242	225	213	204	
	Edh at Noon	36	44	54	67	64	60	59	55	49	43	35	33	
All-Sky Solar Radiation	RadAvg	445	777	1194	1598	1985	2260	2141	1967	1565	1026	618	399	
	RadStd	71	99	108	156	129	95	91	97	91	81	73	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.80	-1.93	-1.76	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page